

CHAPTER 7

CONCLUSIONS

This Chapter is organized as follows: Section 7.1 presents the summary of this thesis. Section 7.2 explains the applications of the proposed methodologies and Section 7.2 presents future research directions for this research work.

7.1 SUMMARY OF THESIS

The research work presented here provides a series of novel approaches for WQM and WQP systems. The use of virtual sensors combined with DL models has been thoroughly explored, resulting in a set of models that replace the traditional WQM systems. The first part of the research work introduced the DL Enhanced Virtual Sensing (DLEVS) mechanism, a novel approach for monitoring DO levels that combines the benefits of LSTM DL-RNN with the strengths of Virtual Sensing (VS) techniques. The model proved effective in providing accurate and reliable estimations of DO levels, demonstrating the promising potential of DL models in virtual sensing, especially in WQM. In the second part of the work, a detailed evaluation of standalone DL models was carried out, identifying their strengths and limitations. Through the evaluation, it was found that none of the models significantly outperformed the others across all WQ parameters for different datasets, highlighting the need for more sophisticated models. This laid the groundwork for the introduction of hybrid models and attention based models, which leverage the strengths of individual models to achieve superior predictive performance.

The third part of the research provides an introduction of hybrid CNN-LSTM and CNN-GRU models, which combined the strengths of LSTM and GRU models with the robust feature extraction capabilities of Convolutional Neural Networks (CNNs). The hybrid models demonstrated superior performance in terms of prediction accuracy and computation efficiency, proving their efficacy in the realm of WQP. Finally, in the fourth part of the research, attention-based models, namely ABi-LSTM and ABi-GRU, were introduced and tested. These models integrated the advantages of LSTM, GRU, and bidirectional models with the ability of an attention mechanism to focus on crucial data by assigning different weights. The attention-based models achieved optimal results in WQP tasks, significantly enhancing the accuracy of prediction and computation efficiency. The research has successfully introduced and validated several

novel models for WQM and WQP. These models could be used as a tool to enhance the efficiency and reliability of monitoring systems in such environments.

7.2 FUTURE WORK

The research presented in this thesis opens up several potential avenues for future work in the field of WQ management. Despite the fact that the offered methodologies have demonstrated improvement over prior benchmarks, there is still scope for improvement based on these new viewpoints to attain more efficiency and reliability. The following are some future research directions that can be pursued to further enhance the methodologies and applications discussed in this thesis:

- Investigate the application of transfer learning techniques in the context of WQM and WQP. Transfer learning can leverage pre-trained DL models on large-scale datasets to improve the performance of models in scenarios with limited labeled data. By fine-tuning pre-trained models or using them as feature extractors, it may be possible to enhance the accuracy and generalization capabilities of the proposed methodologies.
- Explore the integration of other environmental parameters and factors into the DL models. WQ in aquaculture systems is influenced by various factors such as pH, turbidity, ammonia levels, and nutrient concentrations. Investigate the feasibility of incorporating these additional parameters into the DL models to provide a more comprehensive and holistic assessment of WQ.
- Conduct further research on data augmentation techniques specifically tailored for WQ data. Data augmentation can help alleviate data scarcity issues and improve the robustness of DL models. Investigate methods to generate realistic synthetic data that capture the characteristics of different WQ conditions, thereby expanding the training dataset and enhancing model performance.

Exploring these future research directions will help in advancing the field of DL-based WQM and WQP, ultimately contributing to more efficient, accurate, and sustainable aquaculture WQ management practices.